

Tran Quoc Truong

Applied AI Engineer · AI Agents & RAG · Backend Systems

📍 Ho Chi Minh City, Vietnam ✉ tranquoctruong20@gmail.com 🌐 portfolio.truong51972.id.vn 📄 truong51972
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Summary

Applied AI Engineer building agentic applications, RAG and retrieval systems, LLM evaluation infrastructure, and Python backend platforms, with a focus on reliability, observability, and practical delivery.

Experience spans enterprise code intelligence, trace-based AI assurance, model fine-tuning and serving, quality engineering, and CI-driven delivery; independently develops Omni-Agent as a modular platform for knowledge, agents, and evaluation.

Technical Skills

AI & Agentic Systems: RAG, Hybrid Retrieval, LangGraph, MCP (Model Context Protocol), Tool Calling, Context Engineering, LiteLLM, Qdrant, pgvector, Docling

LLM Evaluation & Observability: LLM Evaluation, LLM-as-a-Judge, OpenTelemetry (OTel), Runtime Trace Reconstruction, Regression Testing, Deterministic Quality Gates, Langfuse

Backend Engineering: Python, FastAPI, Django/DRF, Pydantic, SQLAlchemy, PostgreSQL, Redis, Celery, SSE / Streaming APIs, Pytest, SQL, React, TypeScript

ML, Fine-Tuning & Model Serving: PyTorch, Hugging Face Transformers, PEFT, Unsloth, LoRA/QLoRA, vLLM, BitsAndBytes Quantization, Synthetic Data Generation

Infrastructure & Delivery: Docker, Docker Compose, Kubernetes, Google Cloud Platform, RunPod, Nginx, MinIO, GitLab CI/CD, Azure DevOps Pipelines, Linux

Experience

FPT Software , Applied AI Engineer	Ho Chi Minh City, Vietnam Dec 2024 – present 1 year 11 months
IQP (Code Intelligence & Retrieval): Built hybrid code retrieval combining dense search, BM25, and code-graph relationships across enterprise repositories, exposing repository-aware engineering context to AI clients through Model Context Protocol (MCP). Agent Assurance / Flezi Polaris: Architected trace-based evaluation for agentic applications using LiteLLM and OpenTelemetry capture, multi-turn trajectory reconstruction, behavioral evaluation, and reproducible regression gates; the solution received the Golden Solution Prize (1st place) at IVS Solution Day 2.0 2026. Automation & Delivery Engineering: Developed automated validation across API, database, web, and mobile layers and integrated repeatable execution into GitLab and Azure DevOps CI pipelines. Banking Systems Verification: Built reusable cross-platform automation and SQL-based validation across banking and payment systems, reducing manual regression effort by approximately 70%; an omnichannel automation proof of concept contributed to approximately 20 person-months of follow-on delivery work.	
FPT Software Academy , Full-Stack Developer Intern	Ho Chi Minh City, Vietnam Sept 2024 – Dec 2024 4 months
- Built the backend for an online learning and assessment platform using Django with role-based access control and automated exam workflows. - Integrated a FastAPI computer-vision service for AI-assisted real-time exam proctoring and face detection. - Containerized the multi-service architecture with Docker Compose, Nginx, and Cloudflare Tunnel.	

Achievements

Golden Solution Prize (1st Place) — IVS Solution Day 2.0 2026 Awarded the Golden Solution Prize (1st place) on September 18, 2026 for Flezi Polaris, an Agent Assurance solution focused on AI agent evaluation, trace-based evidence capture, and governed quality verification.	Sept 2026
2nd Place — IVS Hackathon 2026 Awarded 2nd place for independently architecting an AI Agent Verification engine featuring runtime evidence capture, trace-based behavioral evaluation, and governed quality gates; subsequently integrated into Omni-Agent.	Aug 2026
Second Prize — Autonomous Driving Research Paper Competition Co-authored research combining YOLOv8 perception, lane segmentation, and PID steering control in Unity simulation, completing the autonomous benchmark in 125.8 seconds with a perfect 100% score.	Apr 2024

Featured Projects

Omni-Agent — Applied AI Platform (Knowledge Builder & Agent Assurance) A modular Applied AI platform combining versioned document knowledge pipelines, agent runtimes, and agent evaluation within a project-centric multi-service architecture. - Platform Architecture: Designed a modular multi-service architecture with project-scoped control plane, isolated domain services, asynchronous workers, PostgreSQL, Redis, S3-compatible object storage, and containerized deployment. - Knowledge Builder: Built a versioned document pipeline covering source ingestion, Docling parsing/OCR, chunking, Qdrant indexing, citation-grounded retrieval, and MCP-based knowledge access. - Agent Runtime & Integration: Implemented LangGraph-based agent workflows with tool calling, context-aware retrieval, MCP integration, and SSE streaming for multi-turn execution over project knowledge. - Agent Assurance: Integrated trace-based evidence capture, evaluation, regression, and quality-gating capabilities into the platform.
APIT — Agent Programmatic Integration Testing AI-assisted API testing framework generating structured test scenarios from API documentation via fine-tuned LLMs. - Developed a bilingual synthetic dataset generation pipeline producing 1,258 domain-specific API testing samples. - Fine-tuned Qwen2.5-3B with Unsloth and PEFT LoRA/QLoRA, achieving macro-F1 of 0.655 (+41.5% relative improvement over Llama-3.2-3B baseline). - Deployed the fine-tuned model on RunPod through an OpenAI-compatible vLLM service with BitsAndBytes quantization and runtime LoRA adapters. - Integrated pgvector-backed cosine-similarity retrieval and a hybrid evaluation pipeline combining LLM-as-a-Judge, fuzzy matching, and deterministic JSON-schema validation.

E-Commerce AI Assistant — Conversational RAG & Search End-to-end shopping assistant combining filtered vector search and multi-agent conversational retrieval. - Built product semantic search using Milvus, cosine similarity, embeddings, and dynamic category/price filters. - Implemented multi-turn conversational agent with LangGraph and LangChain for context-aware product consultation. - Containerized multi-service architecture with FastAPI, PostgreSQL, Redis, Milvus, and Docker Compose.
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Education

FPT School of Business & Technology , Master of Software Engineering in AI	Ho Chi Minh City, Vietnam May 2026 – present
FPT University , Bachelor of Artificial Intelligence	Ho Chi Minh City, Vietnam 2021 – 2025

Languages

Vietnamese: Native

English: Professional working proficiency